PIZZA SALES SQL FINDINGS

Analyzed a pizza sales dataset using SQL to extract business insights. Performed basic to advanced queries including revenue calculation, order distribution analysis, category-wise performance, and cumulative revenue tracking. Applied joins, aggregation, grouping, and window functions to evaluate sales trends and top-performing products.

Task list

- Retrieve the total number of orders placed.
- Calculate the total revenue generated from pizza sales.
- Join the necessary tables to find the total quantity of each pizza category ordered.
- Determine the distribution of orders by hour of the day.
- Join relevant tables to find the category-wise distribution of pizzas.
- Group the orders by date and calculate the average number of pizzas ordered per day.
- Determine the top 3 most ordered pizza types based on revenue.
- Calculate the percentage contribution of each pizza type to total revenue.
- Analyze the cumulative revenue generated over time.
- Determine the top 3 most ordered pizza types based on revenue for each pizza category.

Retrieve the total number of orders placed.

```sql
SELECT
    COUNT(order_id) AS Total_Orders
FROM
    order_details;
```

Result Grid | Filter Rows: | Export:

	Total_Orders
▶	48620

Calculate the total revenue generated from pizza sales.

```sql
SELECT
    ROUND(SUM(od.quantity * p.price),2) AS Total_REVENUE
FROM
    order_details AS od
        INNER JOIN
    pizzas AS p ON od.pizza_id = p.pizza_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

	Total_REVENUE
▸	817860.05

Identify the highest-priced pizza.

```sql
SELECT
    pt.name, p.price
FROM
    pizza_types AS pt
        INNER JOIN
    pizzas AS p ON pt.pizza_type_id = p.pizza_type_id
ORDER BY price DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	price
The Greek Pizza	35.95

Identify the most common pizza size ordered.

```sql
SELECT
    p.size, COUNT(od.quantity) AS Quantity
FROM
    pizzas AS p
        INNER JOIN
    order_details AS od ON p.pizza_id = od.pizza_id
GROUP BY p.size
ORDER BY Quantity DESC;
```

size	Quantity
L	18526
M	15385
S	14137
XL	544
XXL	28

List the top 5 most ordered pizza types along with their quantities.

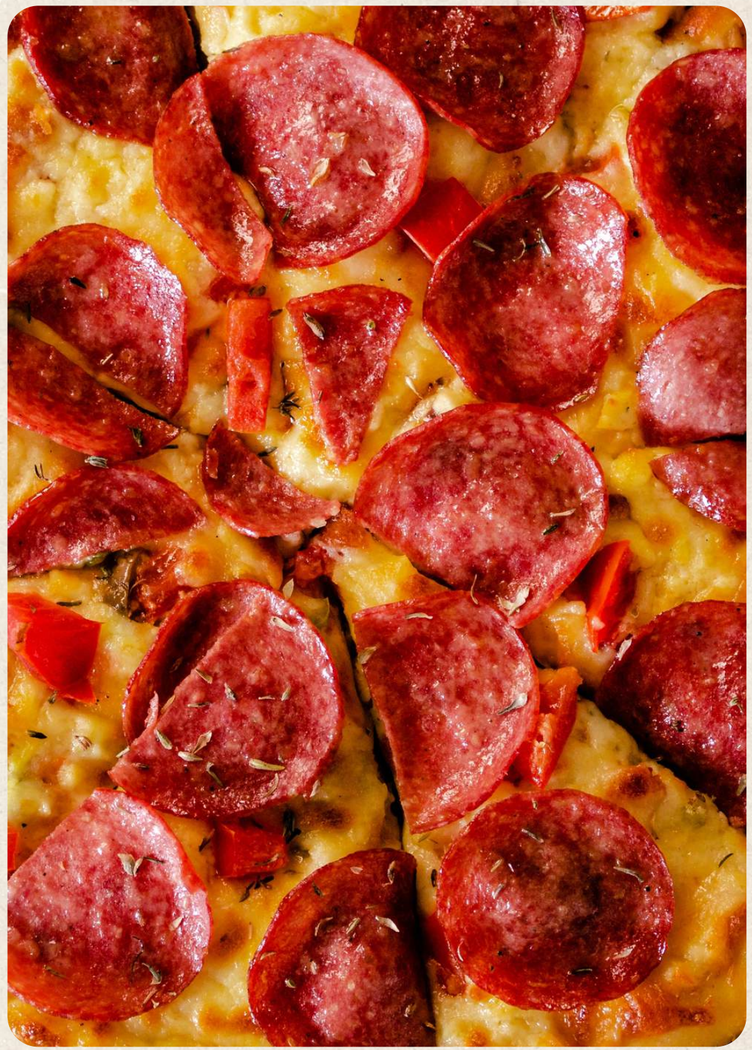
```
SELECT
    pt.name, SUM(od.quantity)
FROM
    pizza_types AS pt
        INNER JOIN
    pizzas ON pt.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details AS od ON od.pizza_id = pizzas.pizza_id
GROUP BY pt.name
ORDER BY SUM(od.quantity) DESC
LIMIT 5;
```

name	SUM(od.quantity)
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Join the necessary tables to find the total quantity of each pizza category ordered.

```sql
SELECT
    pt.category, SUM(od.quantity) Total_Quantity
FROM
    pizza_types AS pt
        INNER JOIN
    pizzas ON pt.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details AS od ON od.pizza_id = pizzas.pizza_id
GROUP BY pt.category
ORDER BY Total_Quantity DESC;
```

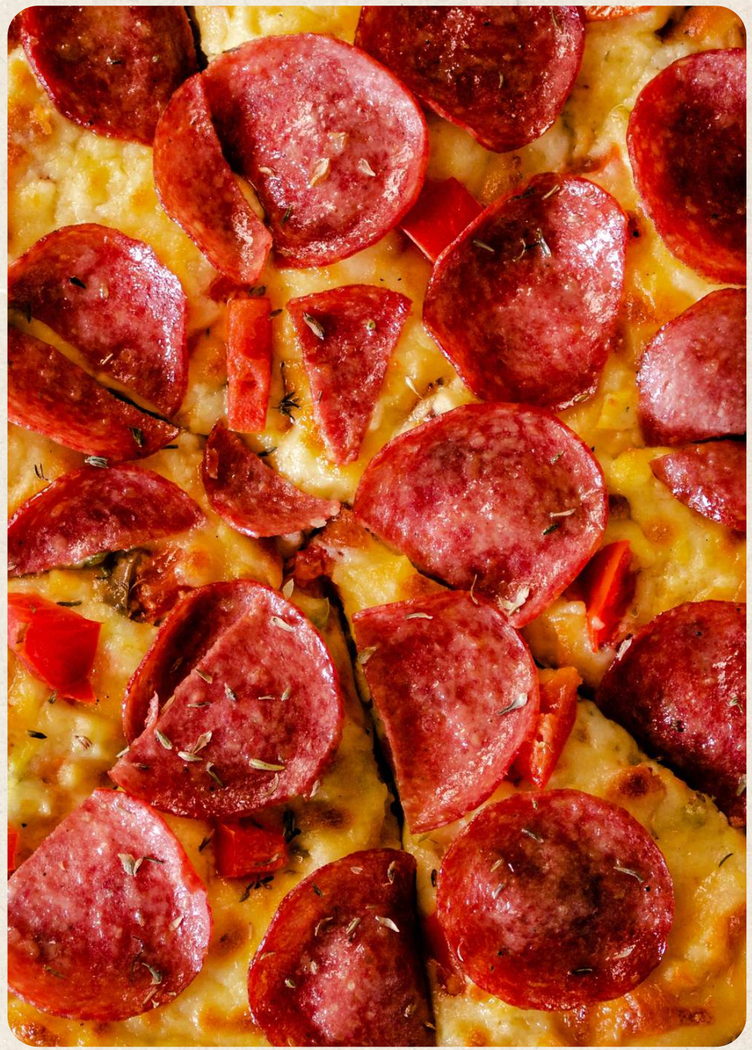
category	Total_Quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050



Determine the distribution of orders by hour of the day.

```sql
SELECT
    HOUR(order_time) AS Hour_Day, COUNT(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time)
ORDER BY COUNT(order_id) DESC;
```

Hour_Day	order_count
12	2520
13	2455
18	2399
17	2336
19	2009
16	1920
20	1642
14	1472

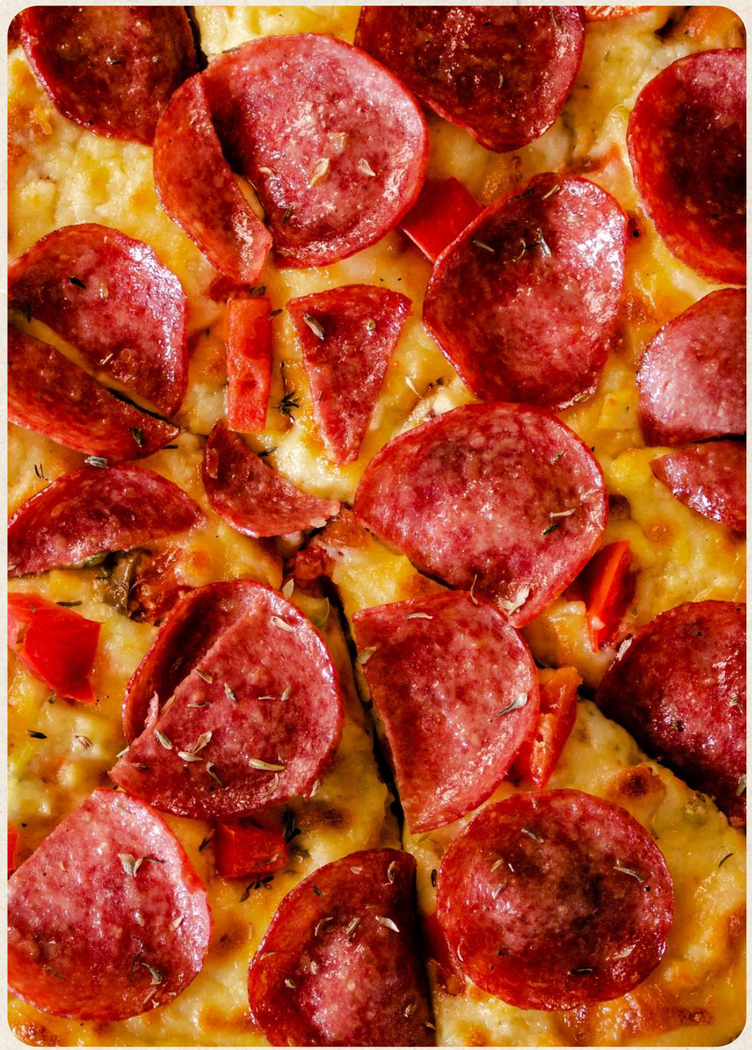


Join relevant tables to find the category-wise distribution of pizzas.

```
select name, category from pizza_types;


SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category
ORDER BY COUNT(name) DESC;
```

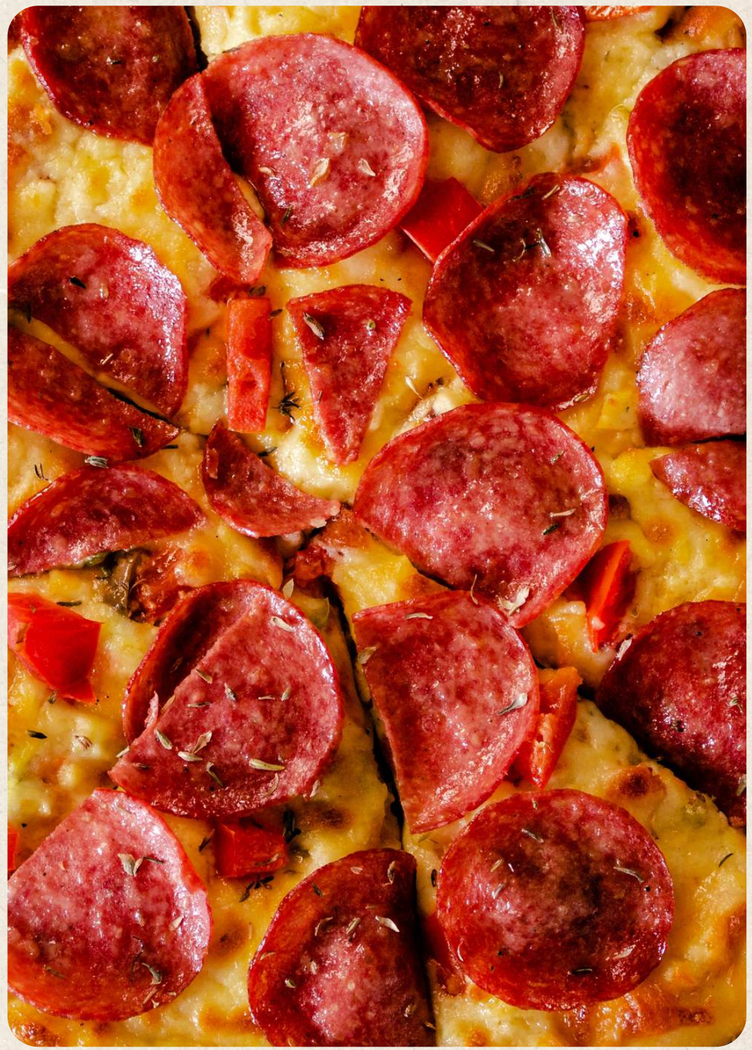
category	COUNT(name)
Supreme	9
Veggie	9
Classic	8
Chicken	6



Group the orders by date and calculate the average number of pizzas ordered per day.

```sql
SELECT
    AVG(Order_Per_Day)
FROM
    (SELECT
        o.order_date, SUM(od.quantity) AS Order_Per_Day
    FROM
        orders AS o
    INNER JOIN order_details AS od ON o.order_id = od.order_id
    GROUP BY o.order_date) AS order_Quantity;
```

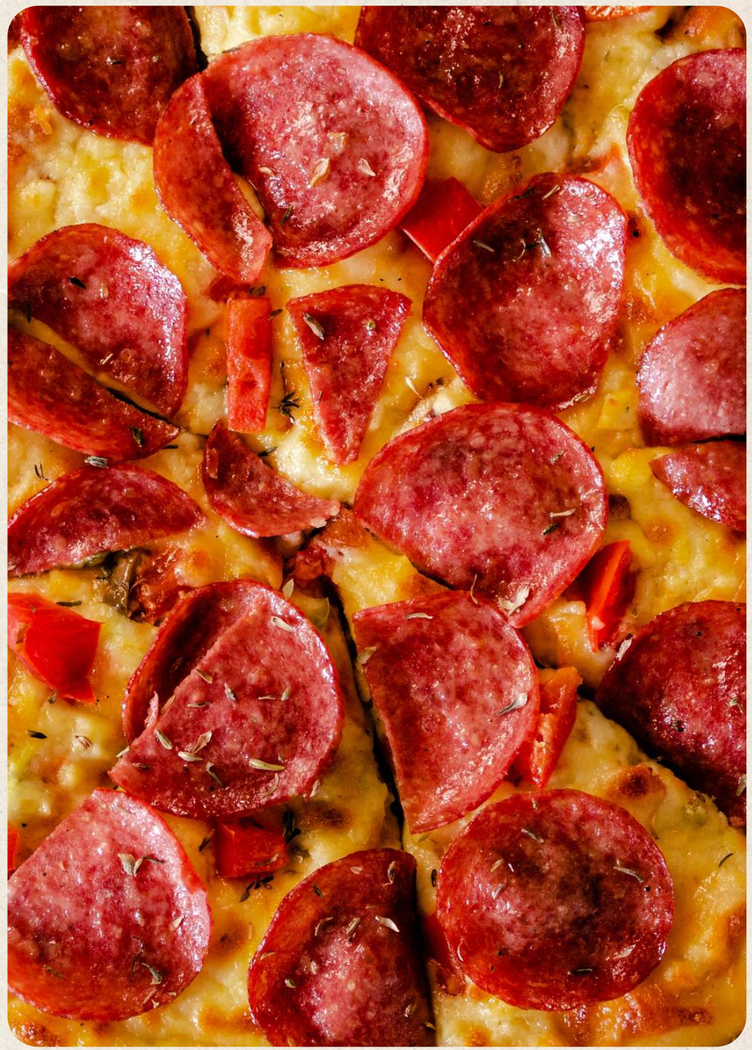
	AVG(Order_Per_Day)
▸	138.4749



Determine the top 3 most ordered pizza types based on revenue.

```sql
SELECT
    pt.name, SUM(od.quantity * p.price) AS Revenue
FROM
    pizza_types AS pt
        INNER JOIN
    pizzas AS p ON pt.pizza_type_id = p.pizza_type_id
        JOIN
    order_details AS od ON od.pizza_id = p.pizza_id
GROUP BY pt.name
ORDER BY Revenue DESC
LIMIT 3
```

name	Revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5



Calculate the percentage contribution of each pizza type to total revenue.

```sql
SELECT
    pt.category,
    Round(SUM(od.quantity * p.price) / (SELECT
            ROUND(SUM(od.quantity * p.price), 2) AS Total_REVENUE
        FROM
            order_details AS od
                INNER JOIN
            pizzas AS p ON od.pizza_id = p.pizza_id) * 100,2) AS Revenue
FROM
    pizza_types AS pt
        INNER JOIN
    pizzas AS p ON pt.pizza_type_id = p.pizza_type_id
        JOIN
    order_details AS od ON od.pizza_id = p.pizza_id
GROUP BY pt.category
ORDER BY Revenue
```

category	Revenue
Veggie	23.68
Chicken	23.96
Supreme	25.46
Classic	26.91

Analyze the cumulative revenue generated over time.

```
select order_date, revenue, round(sum(Revenue) over(order by order_date),2) as cum_revenue
from
(select o.order_date, sum(od.quantity * p.price) as Revenue
from order_details as od inner join pizzas as p
on od.pizza_id = p.pizza_id
join orders as o
on o.order_id = od.order_id group by o.order_date) as Sales;
```

order_date	revenue	cum_revenue
2015-01-01	2713.8500000000004	2713.85
2015-01-02	2731.8999999999996	5445.75
2015-01-03	2662.3999999999996	8108.15
2015-01-04	1755.4500000000003	9863.6
2015-01-05	2065.95	11929.55
2015-01-06	2428.95	14358.5
2015-01-07	2202.2000000000003	16560.7
2015-01-08	2838.3499999999995	19399.05
2015-01-09	2127.3500000000004	21526.4
2015-01-10	2463.95	23990.35

Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
select * from
(select category, name, revenue, rank()over(partition by category order by revenue desc) as rn from
(SELECT
    pt.category, pt.name, SUM(od.quantity * p.price) as revenue
FROM
    pizza_types AS pt
        INNER JOIN
    pizzas AS p ON pt.pizza_type_id = p.pizza_type_id
        JOIN
    order_details AS od ON od.pizza_id = p.pizza_id
GROUP BY pt.category, pt.name) as a) as b
 where rn <= 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	name	revenue	rn
Chicken	The California Chicken Pizza	41409.5	3
Classic	The Classic Deluxe Pizza	38180.5	1
Classic	The Hawaiian Pizza	32273.25	2
Classic	The Pepperoni Pizza	30161.75	3
Supreme	The Spicy Italian Pizza	34831.25	1
Supreme	The Italian Supreme Pizza	33476.75	2
Supreme	The Sicilian Pizza	30940.5	3
Veggie	The Four Cheese Pizza	32265.70000000065	1

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